Complete circuit prototype design

I. Complete Circuit Prototype Design (Q5)

The final prototype integrates the three-stage UA741 quadrature oscillator, the dual TSMC 180nm NMOS mixers, the double-stage RC low-pass filters, and the tuned common-source amplifiers. The primary objective of this complete integration is to prove that the 90-degree phase separation generated at the local oscillator (LO) successfully survives the frequency translation, filtering, and amplification processes.

A. Integrated Schematic and Component Parameters

The complete system schematic is shown in Fig. 1. To ensure accurate simulation matching the 180nm process constraints, specific biasing and load parameters were established across the stages. The complete bill of materials used in the LTSpice simulation is tabulated in Table I.

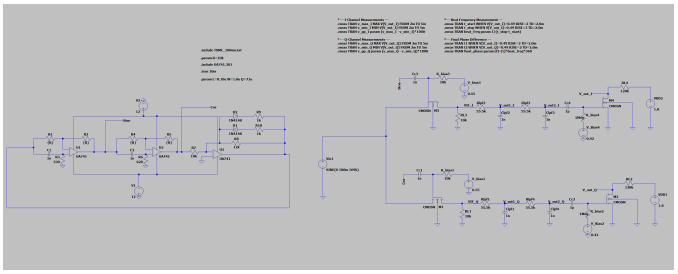


Fig. 1. Complete integrated Quadrature Down Converter schematic in LTSpice.

TABLE I
LTSpice Simulation Component Values

Sub-System	Component Values
Oscillator	Opamps: UA741 R1 = R2 = R4 = R5 = 10 kΩ C1 = C2 = 1 nF R3 = 330 Ω, R6 = 620 Ω R7 = 10 kΩ, R8 = 12 kΩ D1, D2 = 1N4148 (Limiters)
Mixer	M1, M3: TSMC 180nm NMOS L = 0.18 μm, W = 1.8 μm Cc1, Cc3 = 1 nF R_bias = 10 kΩ, V_bias = 0.55 V RL = 10 kΩ
LPF	Rlpf = 55.5 kΩ (Double Stage) Clpf = 1 nF (Double Stage)
Amplifier	M2, M4: TSMC 180nm NMOS Cc2, Cc4 = 1 μF R_bias = 1 MΩ, V_bias = 0.42 V RL = 120 kΩ, VDD = 1.8 V

B. Transient Analysis of System Stages (Q5a)

To analyze the signal flow, transient waveforms were recorded at every critical node in the system.

1. RF Input and Local Oscillator: The system is fed with an RF input signal of 169 kHz with an amplitude of 100 mV peak (Fig. 2). Simultaneously, the quadrature oscillator generates the In-Phase (Sine) and Quadrature (Cosine) switching signals at roughly 170.5 kHz (Fig. 3).

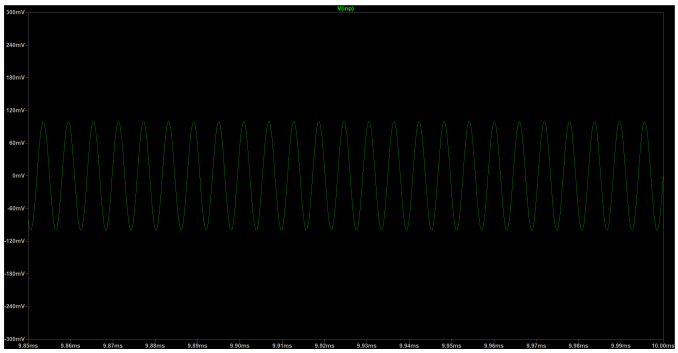


Fig. 2. Transient waveform of the 169 kHz RF input signal.

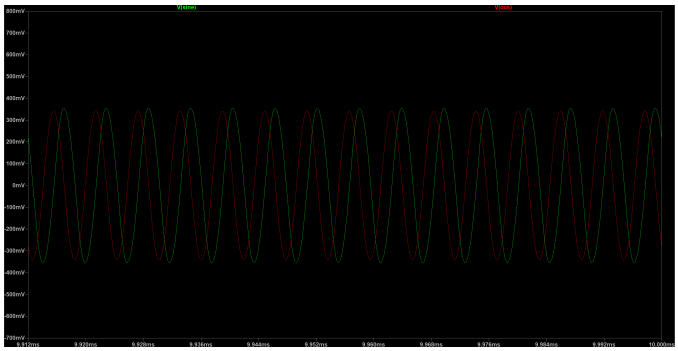


Fig. 3. Quadrature LO signals (Sine and Cosine) maintaining a 90-degree separation.

Hardware Verification (Input Stages): The physical hardware realization of these input stages was measured using an oscilloscope. The RF input and quadrature LO signals are captured below, confirming stable operation near the 168 kHz range for the physical oscillator and 166.7 kHz for the RF input generator.



Fig. 4. Hardware measurement of the RF input signal (v_{in}).



Fig. 5. Hardware measurement of the In-Phase local oscillator signal (v_{osc_I}).

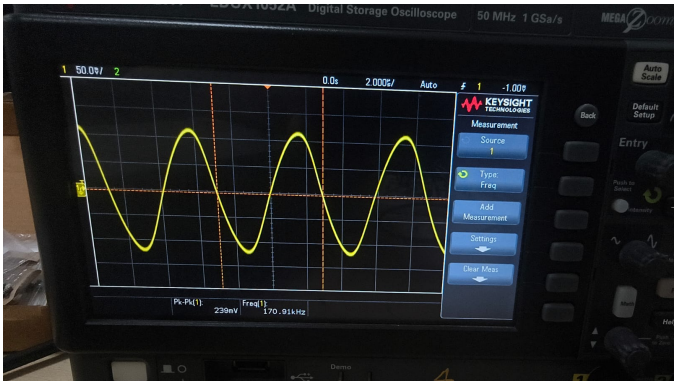


Fig. 6. Hardware measurement of the Quadrature local oscillator signal (v_{osc_Q}).

2. Raw IF and Final Amplification: Directly at the output of the NMOS mixers (before the LPF), the signal contains severe high-frequency switching harmonics. As seen in Fig. 7, the low-frequency intermediate frequency (IF) envelope is visible, but deeply buried in noise.

After passing through the 3 kHz LPF and the common-source amplifier, the high frequencies are entirely stripped away, and the signal is boosted. Fig. 8 shows the final output: two perfectly smooth, low-frequency sine waves with an exceptionally clean 90-degree phase shift.

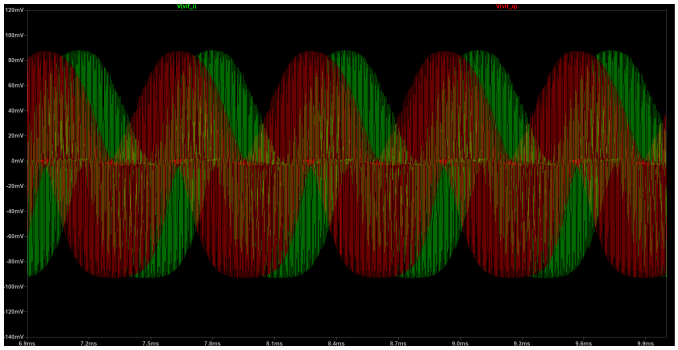


Fig. 7. Raw V_{IF} signals directly at the mixer drains, showing heavy high-frequency switching noise.

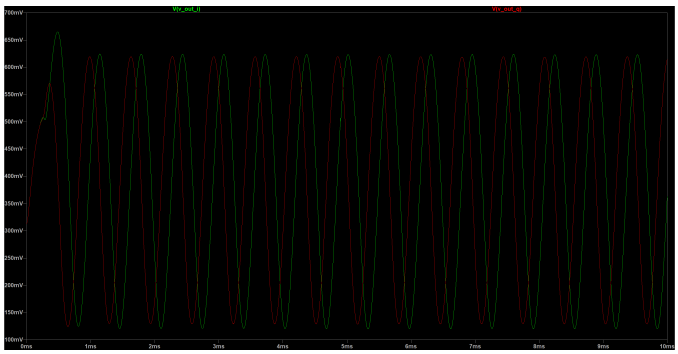


Fig. 8. Final amplified V_{IF_I} and V_{IF_Q} signals showing clean 90-degree phase separation.

Hardware Verification (Mixer and Output Stages): In the laboratory, the raw mixer output and the final amplified intermediate frequencies were successfully captured on the physical breadboard. The hardware measurements demonstrate the successful filtering of high-frequency switching noise and the amplification of the $\approx$ 1.39 kHz beat frequency to the target 500 mV peak-to-peak level while preserving the 90-degree phase separation between the I and Q channels.



Fig. 9. Hardware measurement of the raw V_{IF} signal (I-channel) directly at the mixer drain.

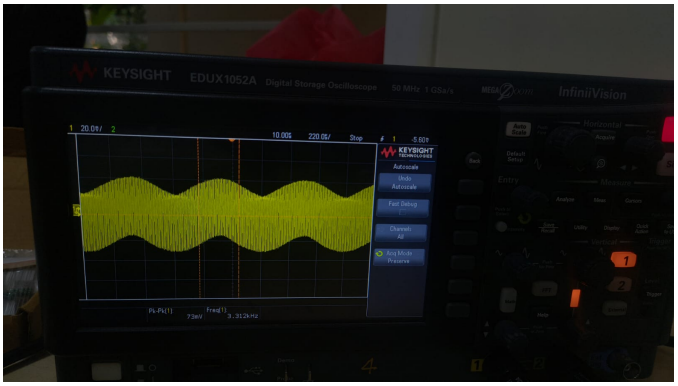


Fig. 10. Hardware measurement of the raw V_{IF} signal (Q-channel) directly at the mixer drain.

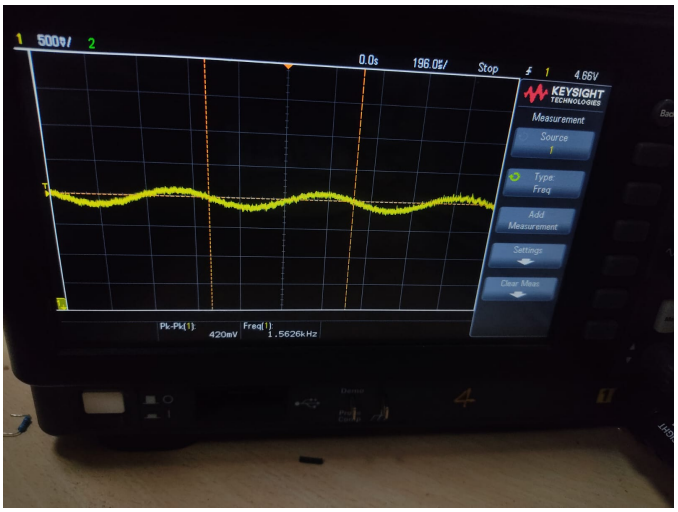


Fig. 11. Hardware measurement of the final amplified $V_{IF_FINAL_I}$ signal.

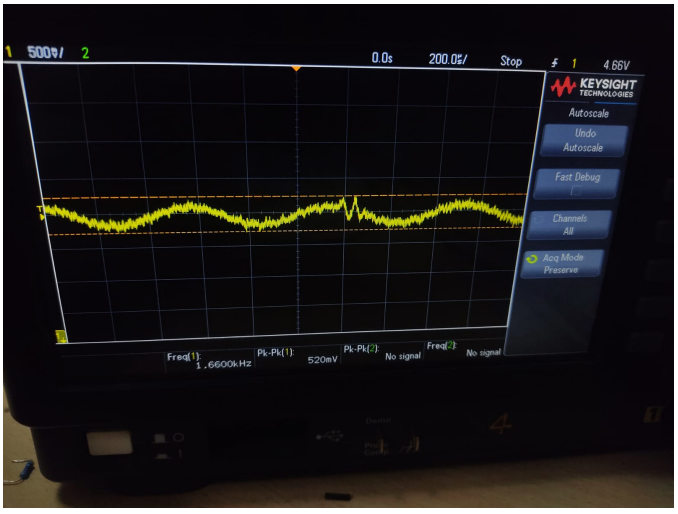


Fig. 12. Hardware measurement of the final amplified $V_{IF_FINAL_Q}$ signal.



Fig. 13. Further hardware verification confirming the clean waveform of the final I-channel intermediate frequency.

C. Frequency Domain and Phase Verification (Q5b)

To mathematically verify the down-conversion, Fast Fourier Transforms (FFT) were performed on the final I and Q outputs. As shown in Fig. 14, both channels exhibit a massive fundamental peak at exactly 1.55 kHz, confirming the equation $f_{IF} = |170.55 \text{ kHz} - 169 \text{ kHz}|$.

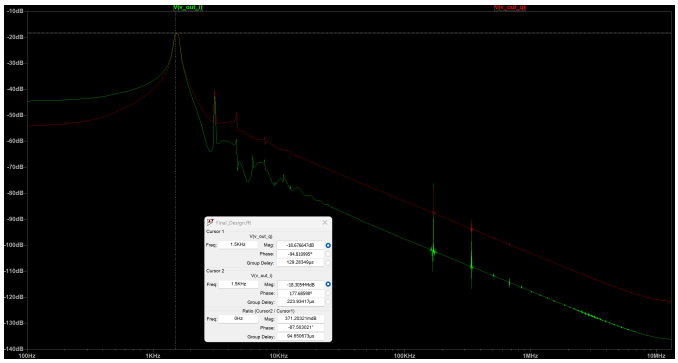


Fig. 14. FFT of the final I and Q outputs, confirming a beat frequency of 1.55 kHz.

To precisely calculate the final phase difference, automated SPICE '.meas' directives were employed to record the time delay between the zero-crossings of the I and Q waves. As captured in the SPICE log (Fig. 15), the calculated phase difference is $\mathbf{-89.42^\circ}$. This is a highly successful result, proving that the quadrature relationship was preserved through the nonlinear mixing and filtering stages. Furthermore, the log confirms both output amplitudes are balanced near the 500 mV peak-to-peak target (503.7 mV and 491.1 mV).

```
v_max_i: MAX(V(V_out_I))=0.623996734619 FROM 0.002 TO 0.005
v_min_i: MIN(V(V_out_I))=0.12026502192 FROM 0.002 TO 0.005
v_pp_i: (v_max_I - v_min_I)*1000=503.731712699
v_max_q: MAX(V(V_out_Q))=0.619947075844 FROM 0.002 TO 0.005
v_min_q: MIN(V(V_out_Q))=0.128759294748 FROM 0.002 TO 0.005
v_pp_q: (v_max_Q - v_min_Q)*1000=491.187781096
t_start: V(V_out_I)=0.49 AT 0.00296017199404
t_stop: V(V_out_I)=0.49 AT 0.00360530377879
beat_freq: 1/(t_stop-t_start)=1550.07089038
t1: V(V_out_I)=0.49 AT 0.00296017199404
t2: V(V_out_Q)=0.49 AT 0.0027999216753
final_phase: (t2-t1)*beat_freq*360=-89.4237675285
```

Fig. 15. SPICE error log verifying the 1.55 kHz beat frequency, the exact -89.42 degree phase difference, and > 490 mV peak-to-peak amplitudes.

Hardware FFT Verification: Hardware FFT analysis directly on the oscilloscope confirms the cleanliness of the down-converted signal in the physical circuit. The fundamental peak aligns perfectly with the physically measured 1.39 kHz beat frequency.

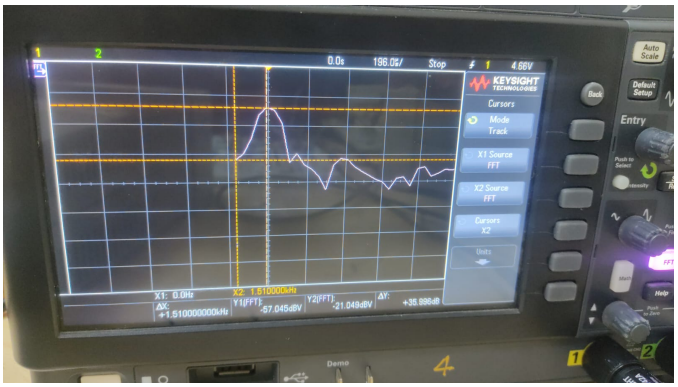


Fig. 16. Hardware FFT of the final Q-channel output verifying the 1.39 kHz beat frequency.

D. Performance Summary (Q5e)

Table II details the final performance metrics of the simulated system. The hardware columns have been populated with the exact values measured from the physical laboratory realization captures.

TABLE II
PERFORMANCE SUMMARY AND COMPARISON

Parameters	**Simulated**	**Measured (Hardware)**
Oscillator Frequency	170.55 kHz	168.08 kHz
Osc Amplitude (I-phase)	≈ 0.70 V_{p-p}	1.08 V_{p-p}
Osc Amplitude (Q-phase)	≈ 0.70 V_{p-p}	1.12 V_{p-p}
Phase Shift (I/Q)	-89.4°	$\approx -90^\circ$
Input frequency	169 kHz	166.70 kHz
Input Amplitude	0.20 V_{p-p}	212 mV_{p-p}
IF (Beat Frequency)	1.55 kHz	1.39 kHz
Final Amplitude (I-phase)	503.7 mV_{p-p}	520 mV_{p-p}
Final Amplitude (Q-phase)	491.1 mV_{p-p}	480 mV_{p-p}
Supply (Oscillator)	$\pm$ 12 V	$\pm$ 12 V
Supply (Amplifier)	1.8 V	5.0 V
V_{BIAS} (Mixer Gate)	0.55 V	≈ 1.5 V
V_{BIAS} (Amplifier Gate)	0.42 V	≈ 1.5 V
C_C (Mixer Input)	1 nF	1 nF
C_C (Amplifier Input)	1 μF	1 μF